

Supplementary Results for The Fidelity–Safety Gap in Neural Wikidata Constraint Repair

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This supplement reports the detailed results underlying the family-level and diagnostic summaries in the main article. Unless stated otherwise, support is measured on the held-out test split of 143,316 repair instances. All reported values come from the evaluations underlying the main results.

Table S1 Historical-repair F1 and base-statement deletion by primary constraint family. The best-system column compares all nine systems evaluated in the main article. Baseline–DM denotes the Definition Majority baseline; Candidate–SR denotes the Satisfaction Reranker. Arrows indicate the preferred direction.

Primary family	Support	Direct–Passive F1 ↑	Direct–Factor F1 ↑	Best F1 ↑	Best system	Candidate–SR base deletion ↓
conflictWith	15,093	0.7205	0.7088	0.8230	Baseline–DM	0.7662
single	17,958	0.5100	0.5626	0.5626	Direct–Factor	0.9898
type	15,001	0.5242	0.4436	0.6838	Baseline–DM	1.0000
distinct	34,232	0.5733	0.9183	0.9183	Direct–Factor	0.9999
valueType	14,999	0.3524	0.4478	0.4908	Baseline–DM	1.0000
itemRequiresStatement	15,001	0.3497	0.5078	0.7512	Baseline–DM	1.0000
inverse	15,001	0.6980	0.7938	0.9011	Baseline–DM	1.0000
valueRequiresStatement	15,003	0.5920	0.7457	0.7457	Direct–Factor	1.0000
oneOf	1,028	0.6951	0.6932	0.8441	Baseline–DM	1.0000

Table S2 Direct–Factor pre-repair satisfaction-head diagnostics by factor family. Support is the number of factor occurrences included in the probe evaluation; positive rate is the share carrying a positive symbolic satisfaction label. The final row is the unweighted mean across the ten families. Arrows indicate the preferred direction where one exists.

Factor family	Support	Positive rate	F1 ↑	AUROC ↑	AUPRC ↑	ECE ↓
conflictWith	1,843,471	0.7164	0.9612	0.9886	0.9949	0.2431
distinct	431,550	0.9407	0.9994	0.9996	1.0000	0.0626
inverse	42,384	0.3294	0.9577	0.9928	0.9898	0.6553
itemRequiresStatement	705,718	0.9935	0.9971	0.8346	0.9983	0.0040
oneOf	3,331	0.7577	0.8622	0.5841	0.8315	0.0872
single	1,368,781	0.9986	0.9993	0.9171	0.9998	0.0003
symmetric	4,698	0.9453	0.9719	0.8393	0.9854	0.0442
type	752,048	0.9429	0.9889	0.9744	0.9976	0.0350
valueRequiresStatement	20,344	0.9631	0.9812	0.8477	0.9927	0.0375
valueType	466,401	0.8708	0.9822	0.9839	0.9967	0.1055
Unweighted mean		0.8458	0.9701	0.8962	0.9787	0.1275

1 Candidate-system reproducibility

Finite candidate pool.

Each candidate is a six-ID vector $(a_s, a_p, a_o, d_s, d_p, d_o)$, with ID 0 denoting NONE. For each action separately, the construction takes the five highest-logit IDs in each of its subject, predicate, and object slots, enumerates their Cartesian product (at most $5^3 = 125$ triples), assigns each triple

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■ **Table S3** Candidate availability and selection by system. Each rate cell reports availability/selection (A/S) as percentages. “Eligible” restricts the denominator to the 54,234 instances whose primary constraint is checkable and violated before repair; “+ base” additionally requires the base statement to remain present. Candidate–SR shares the frozen Direct–Factor proposal generator, so their availability rates are identical. Arrows indicate that higher rates are preferred under the stated suitability requirements.

System	Mean $ \Gamma_i $	All test A/S $\uparrow$	All test + base A/S $\uparrow$	Eligible A/S $\uparrow$	Eligible + base A/S $\uparrow$
Direct–Factor	42.42	34.51 /23.19	9.78 / 3.62	91.19 /61.28	25.85 / 9.57
Candidate–C	42.42	<u>34.42</u> /21.82	<u>9.63</u> / <u>3.40</u>	<u>90.97</u> /57.66	<u>25.46</u> / <u>8.99</u>
Candidate–DP	42.41	34.35/ <u>28.37</u>	9.41/1.20	90.78/ <u>74.96</u>	24.85/3.16
Candidate–SR	42.42	34.51 / 33.16	9.78 /1.68	91.19 / 87.64	25.85 /4.45

the sum of its three raw slot logits, and retains the first 20 after a stable descending sort. When entity/predicate class masks are supplied, they restrict the corresponding slots; a mask containing fewer than five IDs falls back to the full target vocabulary. Heuristic patterns retain at most the first three distinct, nonzero, non-placeholder IDs per component and at most 30 instantiated additions and 30 instantiated deletions.

Before deduplication, candidates occur in the following order: the historical repair when enabled, heuristic additions, heuristic deletions, proposal additions, and proposal deletions. Out-of-range vectors are discarded, exact six-slot duplicates are removed while retaining their first occurrence, and the first 80 remaining vectors form Γ_i . The historical repair is included only when computing training and validation losses; it can contain both actions, whereas proposal and instantiated heuristic candidates contain at most one. Inference and candidate-oracle evaluation never insert it.

■ **Table S4** Heuristic patterns dispatched by the primary constraint family. Every family first proposes deletion of the focus statement and, when present, the competing statement identified by the `other_*` fields. The table lists the additional patterns.

Primary family	Additional candidate pattern
<code>conflictWith, single</code>	Delete overlapping statements formed from the focus and competing subject, property, and value when their roles coincide.
<code>oneOf</code>	Use the same overlap deletions; add the focus property with a value listed by items (P2305).
<code>distinct</code>	Delete the competing subject’s duplicate statement when it has the same property and value as the focus statement.
<code>type</code>	Add a statement about the focus subject using class (P2308) and relation (P2309); if no relation is supplied, use instance of (P31) or subclass of (P279).
<code>valueType</code>	Apply the <code>type</code> construction to the focus object’s entity rather than the focus subject.
<code>itemRequiresStatement</code>	Form $(s_i, p, *)$ for each required property p listed by property (P2306).
<code>valueRequiresStatement</code>	Form $(o_i, p, *)$ for each required property p listed by property (P2306).
<code>inverse</code>	Add (o_i, p^{-1}, s_i) , using inverse property (P1696), or the focus property when no inverse is supplied.
<code>symmetric</code>	No family-specific pattern beyond the general deletions; this family occurs only as a secondary constraint in the benchmark.

The asterisk in the two dependency templates is a wildcard, not a target-vocabulary ID. Because finite heuristic instantiation requires a concrete subject, property, and object, these templates do not themselves add a candidate; a concrete dependency repair must therefore be supplied by the proposal top- k set.

32 **Candidate-C chooser.**

33 The chooser uses one shared 400-dimensional ID embedding for all six candidate slots and averages
 34 the six resulting vectors. It concatenates this 400-dimensional candidate representation with the
 35 Direct-Factor graph representation and scores it with Linear(800, 400), ReLU, and Linear(400, 1).
 36 There is no chooser-specific dropout. A softmax is applied over the candidates of each instance
 37 during training; inference selects their score argmax. The chooser, graph encoder, and proposal
 38 heads are trained jointly under the protocol in the main article.

39 **Candidate-SR reranker.**

40 The frozen Direct-Factor model supplies the proposal logits, while the reranker includes a sep-
 41 arately initialised graph encoder over the factorised input graph. This encoder has an input
 42 projection followed by one 128-dimensional GIN message-passing block; it uses 128-dimensional
 43 node-ID embeddings and dropout 0.5, without role embeddings, predicate-edge attributes, or
 44 factor pressure. Each of the six candidate IDs is embedded in 64 dimensions. Their concatenation
 45 passes through Linear(384, 128), ReLU, dropout 0.1, Linear(128, 128), and ReLU. Concatenating
 46 that output with the 128-dimensional graph representation gives a 256-dimensional vector, scored
 47 by Linear(256, 128), ReLU, dropout 0.1, and Linear(128, 1). The graph encoder and reranker are
 48 trained jointly with Adam, learning rate 10^{-4} , weight decay 10^{-4} , batch size 64, gradient clipping
 49 0.5, and seed 42. Training lasts at most ten epochs, uses the first 25,000 validation instances,
 50 halves the learning rate after one epoch without validation improvement, stops after two such
 51 epochs, and retains the parameter state with minimum mean validation loss.

52 **Edge cases.**

53 An addition or deletion executes only when all three IDs in that action are nonzero; an incomplete
 54 branch is a symbolic no-op. No explicit no-op candidate is inserted, although proposal logits or
 55 the historical repair included during training and validation can yield one. Exact vectors are
 56 deduplicated before operation resolution, so distinct incomplete vectors can resolve to the same
 57 no-op. Deletion is applied before addition, preserving delete-and-reinsert behaviour. The reported
 58 top- k settings normally make every pool nonempty; an empty pool is treated as a failure rather
 59 than being replaced by a fallback candidate. Exact score ties are resolved by the first candidate
 60 in the stable pool order above.